

2) So far we have

$$N = \frac{gV}{(2\pi)^3} \int d^3\vec{p} \frac{1}{e^{\beta(\epsilon_p - \mu)} + 1}$$

$$= \frac{gV}{(2\pi)^3} (4\pi) \frac{2}{c^3} \int_{mc^2}^{\infty} d\epsilon \frac{\epsilon \sqrt{\epsilon^2 - m^2 c^4}}{e^{\beta(\epsilon - \mu)} + 1}$$

$$n_F(\epsilon) = \frac{1}{e^{\beta(\epsilon - \mu)} + 1}$$

$$n'_F(\epsilon) = -\frac{\beta e^{\beta(\epsilon - \mu)}}{(e^{\beta(\epsilon - \mu)} + 1)^2}$$

$$I[f] = \int_{mc^2}^{\infty} d\epsilon f(\epsilon) n_F(\epsilon)$$

$$F(\epsilon) = \int_{mc^2}^{\epsilon} f(\epsilon') d\epsilon'$$

$$= \underbrace{F(\epsilon) n_F(\epsilon)}_{\substack{0 \\ F(mc^2)=0 \\ n_F(\infty)=0}} \Big|_{mc^2}^{\infty} + \int_{mc^2}^{\infty} d\epsilon F(\epsilon) (-n'_F(\epsilon))$$

$\downarrow$
 $\sim \delta(\epsilon - \mu)$
 at high β (low T)

$$= \int_{-\infty}^{\infty} d\epsilon F(\epsilon) (-n'_F(\epsilon))$$

Taylor expand about μ :

$$F(\epsilon) \approx F(\mu) + \underbrace{F'(\mu)}_{\substack{=0 \\ \text{odd fct.}}} (\epsilon - \mu) + \frac{1}{2} (\epsilon - \mu)^2 F''(\mu) + \dots$$

$$I[f] \approx \int_{-\infty}^{\infty} d\epsilon \left[F(\mu) + \frac{1}{2} (\epsilon - \mu)^2 F''(\mu) + \dots \right] \frac{\beta e^{\beta(\epsilon - \mu)}}{(1 + e^{\beta(\epsilon - \mu)})^2}$$

Shift $\epsilon' \rightarrow \epsilon - \mu$

$$= \int_{-\infty}^{\infty} d\epsilon \left[F(\mu) + \frac{1}{2} \epsilon^2 F''(\mu) + \dots \right] \frac{\beta e^{\beta\epsilon}}{(1 + e^{\beta\epsilon})^2}$$

$$\begin{aligned} x &= \beta\epsilon \\ \frac{dx}{\beta} &= d\epsilon \end{aligned}$$

$$= \int_{-\infty}^{\infty} dx \left[F(\mu) + \frac{1}{2} (\hbar T)^2 x^2 F''(\mu) + \dots \right] \frac{e^x}{(1 + e^x)^2}$$

$$\int_{-\infty}^{\infty} dx \frac{e^x}{(1 + e^x)^2} = 1 \quad \int_{-\infty}^{\infty} dx \frac{x^2 e^x}{(1 + e^x)^2} = \frac{\pi^2}{3}$$

$$= F(\mu) + \frac{\pi^2}{6} (kT)^2 F''(\mu)$$

For this GA) $f(\epsilon) = \epsilon \sqrt{\epsilon^2 - m^2 c^4}$

$$F(\epsilon) = \int_{mc^2}^{\epsilon} d\epsilon' \epsilon' \sqrt{\epsilon'^2 - m^2 c^4}$$

$$= \frac{1}{2} \int_0^{\epsilon^2 - m^2 c^4} du \, u^{\frac{1}{2}}$$

$$= \frac{1}{2} \cdot \frac{2}{3} u^{\frac{3}{2}} \Big|_0^{\epsilon^2 - m^2 c^4}$$

$$= \frac{1}{3} [\epsilon^2 - m^2 c^4]^{\frac{3}{2}}$$

$$\begin{aligned} \hookrightarrow F'(\epsilon) &= \left(\frac{1}{3} (\epsilon^2 - m^2 c^4)^{\frac{3}{2}} \right)' - 2\epsilon \\ &= (\epsilon \sqrt{\epsilon^2 - m^2 c^4})' \\ &= \sqrt{\epsilon^2 - m^2 c^4} + \epsilon \frac{1}{\epsilon} \frac{1}{\sqrt{\epsilon^2 - m^2 c^4}} - 2\epsilon \\ &= \sqrt{\epsilon^2 - m^2 c^4} + \frac{\epsilon^2}{\sqrt{\epsilon^2 - m^2 c^4}} \end{aligned}$$

$$N = \frac{gV}{(2\pi)^3} (4\pi) \underbrace{\frac{2}{c^3} \int_{mc^2}^{\infty} d\epsilon \frac{\epsilon \sqrt{\epsilon^2 - m^2 c^4}}{e^{\beta(\epsilon - \mu)} + 1}}_{F(\mu) + \frac{\pi^2}{6} (kT)^2 F''(\mu)}$$

$$= \frac{gV}{(2\pi)^3} \frac{8\pi}{c^3} \left[\frac{1}{3} [\mu^2 - m^2 c^4]^{\frac{3}{2}} + \frac{\pi^2}{6} (kT)^2 \left(\sqrt{\mu^2 - m^2 c^4} + \frac{\mu^2}{\sqrt{\mu^2 - m^2 c^4}} \right) \right]$$

Let $\mu = \epsilon_F + \delta = \sqrt{p_F^2 c^2 + m^2 c^4} + \delta$ be a perturbation
just above $T=0$

$$= \frac{gV}{\pi^2 c^3} \left[\frac{1}{3} [(\epsilon_F + \delta)^2 - m^2 c^4]^{\frac{3}{2}} \right]$$

M

$$1 - \frac{2}{1+\epsilon} \cos(2\pi f_0 dt) z + \frac{1}{(1+\epsilon)^2} z^2$$

$$= \frac{M}{(z-p)(z-\bar{p})}$$

$$1 = \oint_{|z|=1} \frac{M}{(z-p)(z-\bar{p})} dz$$

$$= 1 - 2\pi i (\text{Res}(f, p) + \text{Res}(f, \bar{p}))$$

$$\frac{A}{(z-p)} + \frac{B}{(z-\bar{p})} = \frac{M}{(z-p)(z-\bar{p})}$$

$$= \frac{Az - A\bar{p} + Bz - Bp}{(z-p)(z-\bar{p})} = \frac{(A+B)z - A\bar{p} - Bp}{(z-p)(z-\bar{p})}$$

$$A+B=0 \Rightarrow A=B$$

$$M = B\bar{p} - Bp$$

$$= B(\bar{p} - p)$$

$$B = \frac{M}{\bar{p} - p}$$

$$\begin{array}{c} \uparrow \\ \text{Res at} \\ z = \bar{p} \end{array}$$

$$A = \frac{M}{p - \bar{p}}$$

$$\begin{array}{c} \uparrow \\ \text{Res at} \\ z = p \end{array}$$

$$1 = \frac{M}{\bar{p} - p} + \frac{M}{p - \bar{p}}$$

$$M = \bar{p} - p = (1+\epsilon) [\cos(\gamma) + i\sin(\gamma) - \cos(\gamma)]$$

$$p = (1+\epsilon) e^{2\pi i f_0 / f_s}$$

$$z^2 - (\bar{p} + p) + (1+\epsilon)^2$$

$$= \frac{1}{(1+\epsilon)^2} \left(\frac{z^2}{(1+\epsilon)^2} - \frac{1}{(1+\epsilon)} (\bar{p} + p) + 1 \right)$$

Normalize $M = \frac{1}{(1+\epsilon)^2}$ so that

$$\epsilon = 0.05 \rightarrow 400$$

$$\epsilon = 0.1 \rightarrow 100$$

Complex linear chirp

$$h(t) = A e^{2\pi i (\Omega t + \gamma t^2)}$$

Ω : initial frequency

γ : chirp rate

$$IF : \frac{\partial}{\partial t} [\Omega t + \gamma t^2]$$

$$= \Omega + 2\gamma t \quad \text{instantaneous frequency}$$

↑
Linear
rate IF

rate of change of
 $f(t)$ as t increases

Analyzed with Linear Chirp Transformation

$$H_L(\Omega, \gamma) = \int_{\mathbb{R}} dt \quad h(t) \quad e^{-2\pi i (\Omega t + \gamma t^2)}$$

↑
Detector signal

$$\Rightarrow h(t) = \int_{\mathbb{R}} d\Omega H_L(\Omega, \gamma) e^{2\pi i (\Omega t + \gamma t^2)}$$

"Essentially a
Fourier transform"

Window the input dataset since $h(t)$ is finite

$$H_J(\gamma, \tau, \Omega) = \int_{\mathbb{R}} dt \quad h(t) \quad g_c(\gamma, t-\tau, \Omega) e^{-2\pi i \Omega t}$$

$$g_c(\gamma, t-\tau, \Omega) \equiv \frac{|\Omega_0 + \mu \Omega|}{\sqrt{2\pi}} \frac{1 + \frac{t-\tau}{|t-\tau|}}{2} e^{-(t-\tau)^2 \left[\frac{(\Omega_0 + \mu \Omega)^2}{2} + 2\pi i \gamma \right]}$$

partial
normalization

Ω_0 : arbitrary base frequency parameter

μ : frequency scaling parameter

Spectrogram generation:

Project $H_J(r, \tau, \Omega)$ along chirp rate axis

$$S_J(\tau, \Omega) = \int_{-L}^L dr H_J(r, \tau, \Omega).$$

Assuming r has a rate $\tau \in [-L, L]$.

Ω : freq. axis : $\omega = 2\pi\Omega$ ang. freq.

ASIDE:

$$H_J(r, \tau, \Omega) = \int_{\mathbb{R}} dt h(t) g_c(r, t - \tau, \Omega) e^{-2\pi i \Omega t}$$

$$= \mathcal{F}\{h(t) \cdot g_c(r, t - \tau, \Omega)\}$$

$$= \tilde{\mathcal{F}}\{h(t)\} * \tilde{\mathcal{F}}\{g_c(r, t - \tau, \Omega)\}$$

$$= \tilde{\mathcal{F}}\{h(t)\} * \tilde{\mathcal{F}}\{g_c(r, t, \Omega)\} e^{-2\pi i \Omega \tau}$$

Convolution + 2 FFT ↑
Rotation in $\mathbb{C}$

$$\int_{-\infty}^{\infty} dt \quad e^{-(t-\tau)^2 \frac{\Omega^2}{2}} \underset{\substack{\uparrow \\ \delta(t-\tau)}}{e^{-2\pi i(t-\tau)^2}}$$

$$\int_{-\infty}^{\infty} dt \quad e^{-\frac{\Omega^2}{2}(t-\tau)^2} = \sqrt{\frac{\pi}{a}}$$

$$a = \frac{\Omega^2}{2} = \sqrt{\frac{2\pi}{\Omega^2}} = \frac{1}{\Omega} \sqrt{2\pi}$$

$$\int_{-\infty}^{\infty} h(t) e^{-2\pi i(\Omega t + \gamma t^2)} dt$$

$$\int_{-\infty}^{\infty} [h(t) e^{-2\pi i \gamma t^2}] e^{-2\pi i \Omega t} dt$$

$$= \mathcal{F} \{ h(t) e^{-2\pi i \gamma t^2} \}$$

$$H_L(\Omega, \gamma) = \mathcal{F} \{ h(t) \} * \mathcal{F} \{ e^{-2\pi i \gamma t^2} \}$$

$$X = \frac{58.074}{\omega} = \frac{N}{60} \cdot 58.074$$

11892

238

$$\frac{25.2255}{60} = 12288$$

